CHALLENGE PROJECT #1

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Problem

Iodine Deficiency/Hashimito's Thyroidisis

Hypothyroidism

Side Effects

Motivation

Previous

- Costly
- Large amount of features (30)
- No more budget to gather more data

VS

Our Model

- Final model only uses top 5 features
- Does not need money to gather more data

RESEARCH QUESTION:

Can we accurately predict whether a patient has hypothyroidism using a supervised model trained on a minimal set of features?

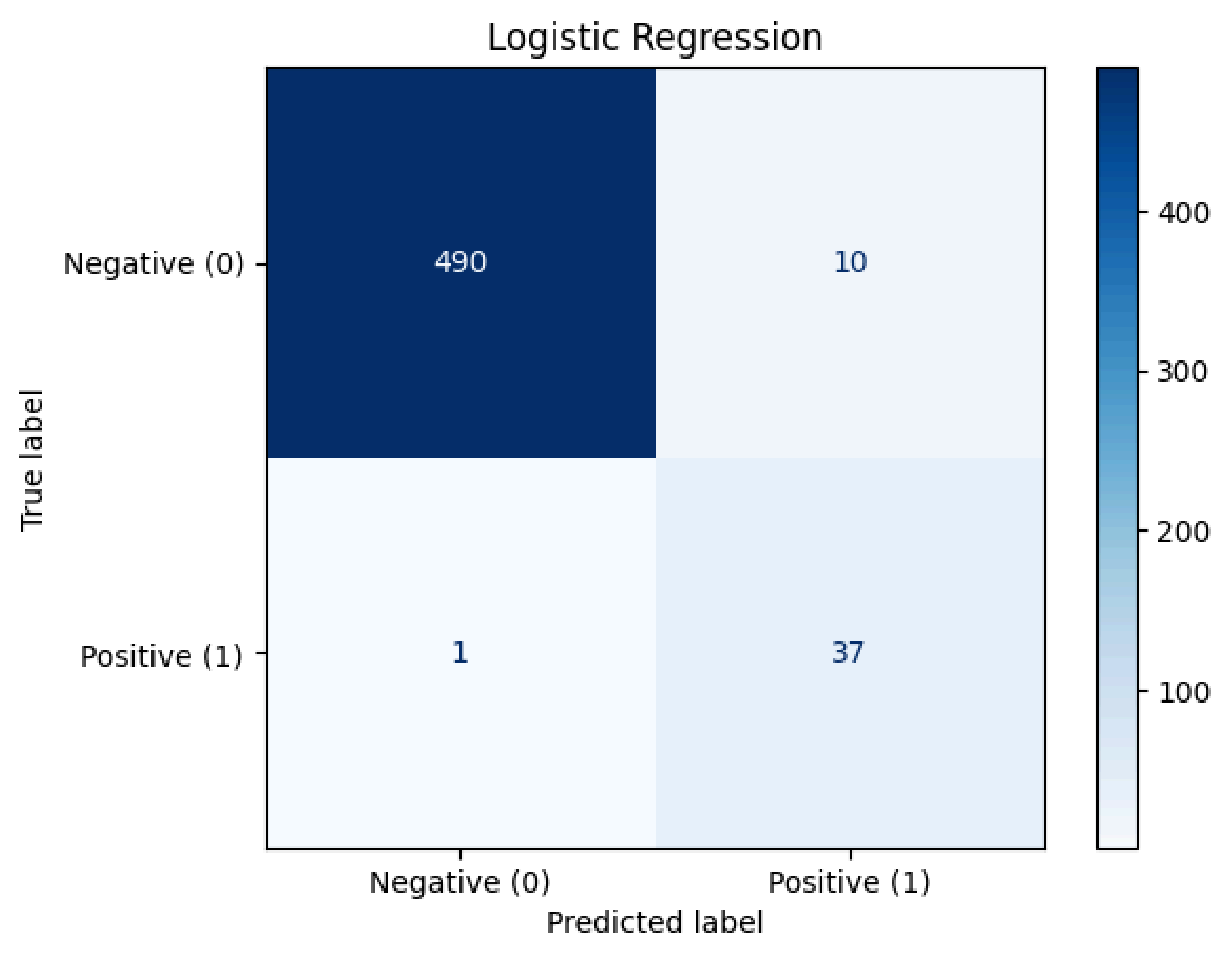
Methodology Overview

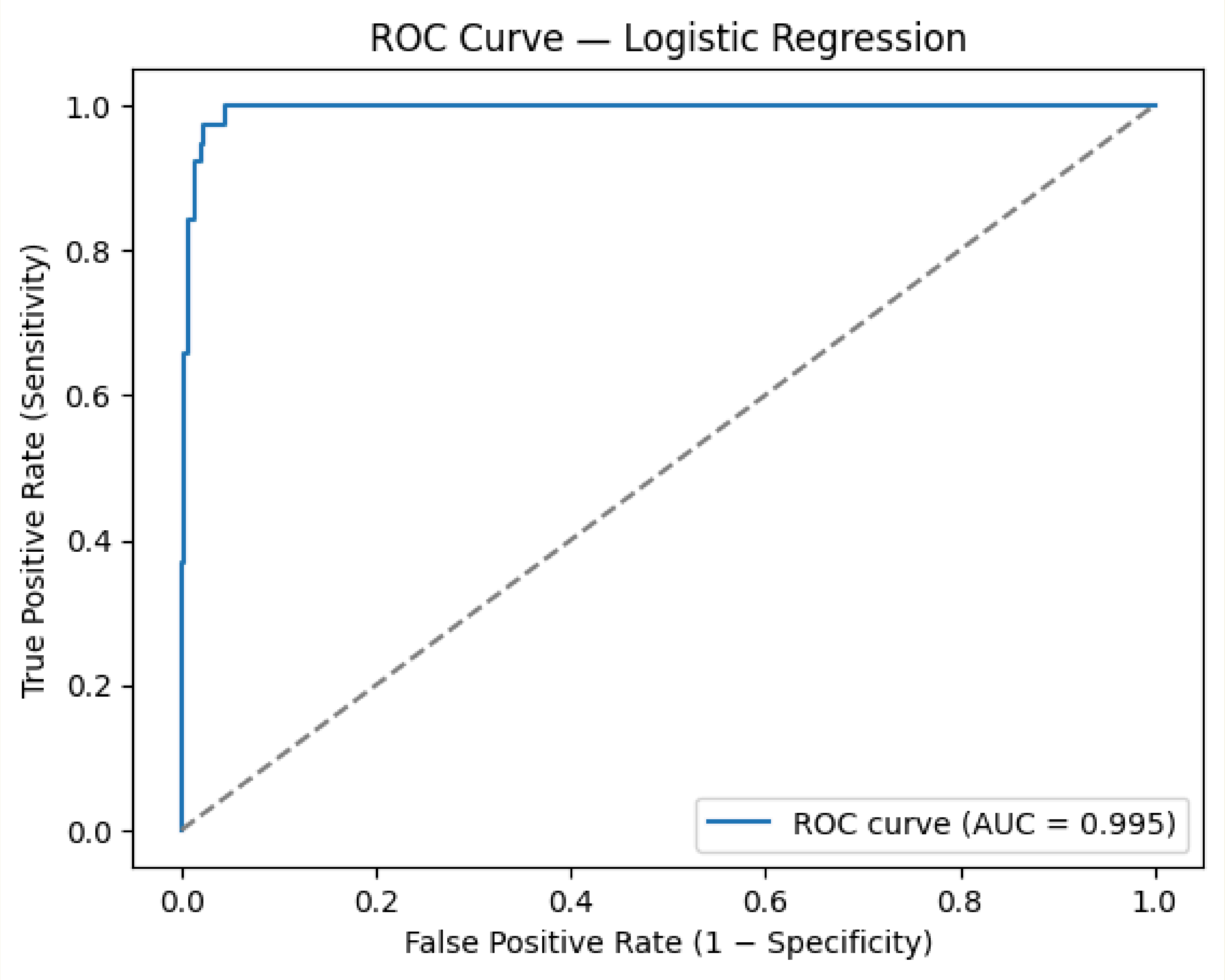
1. Data Cleaning
2. Data analysis
3. Model training and selection
4. Model evaluation
5. Trimming Features

DATA CLEANING

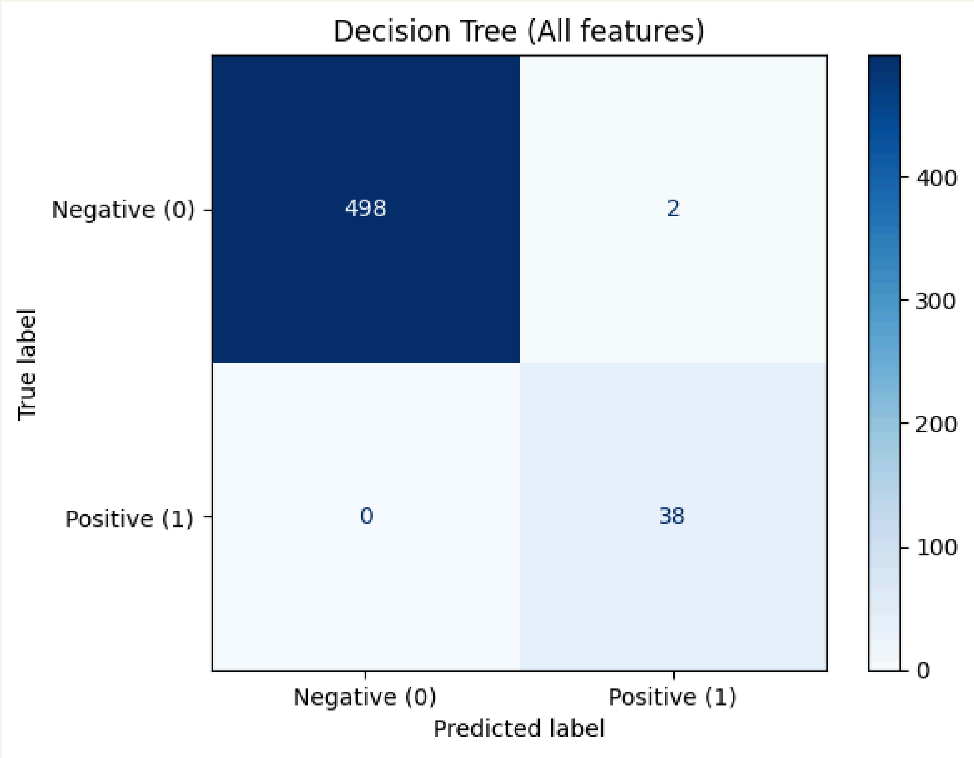
- Separate data into numerical and categorical sets
- replace all missing data in numerical data with median values
- Remove empty TBG column
- Clean ‘Class’ column
- Convert ‘Class’ column into binary values

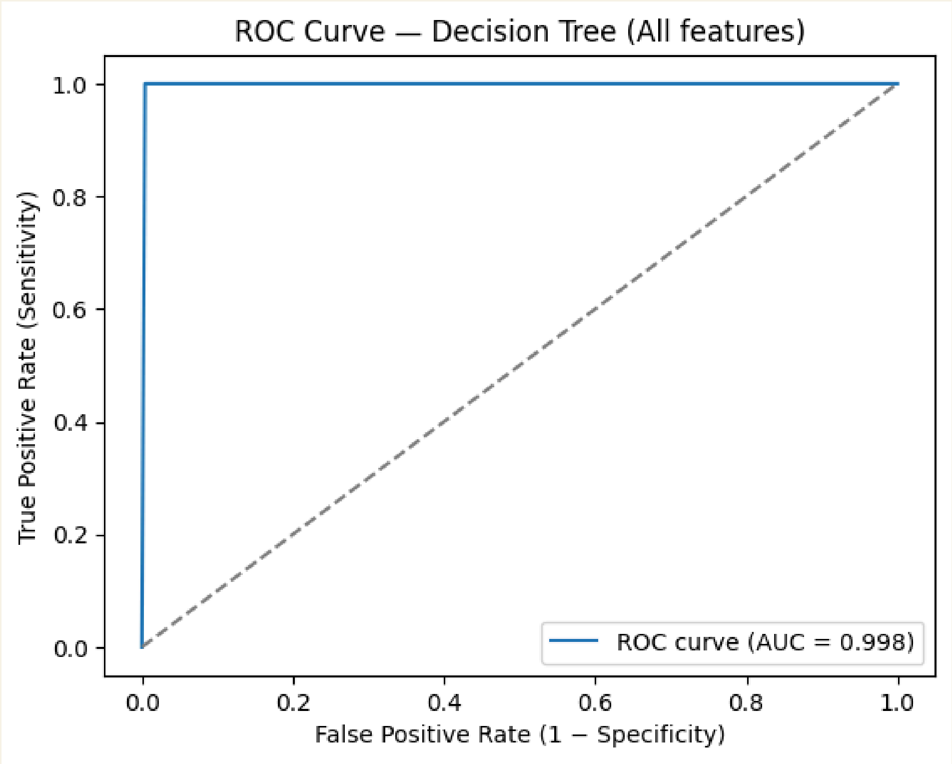
Model: Logistic Regression

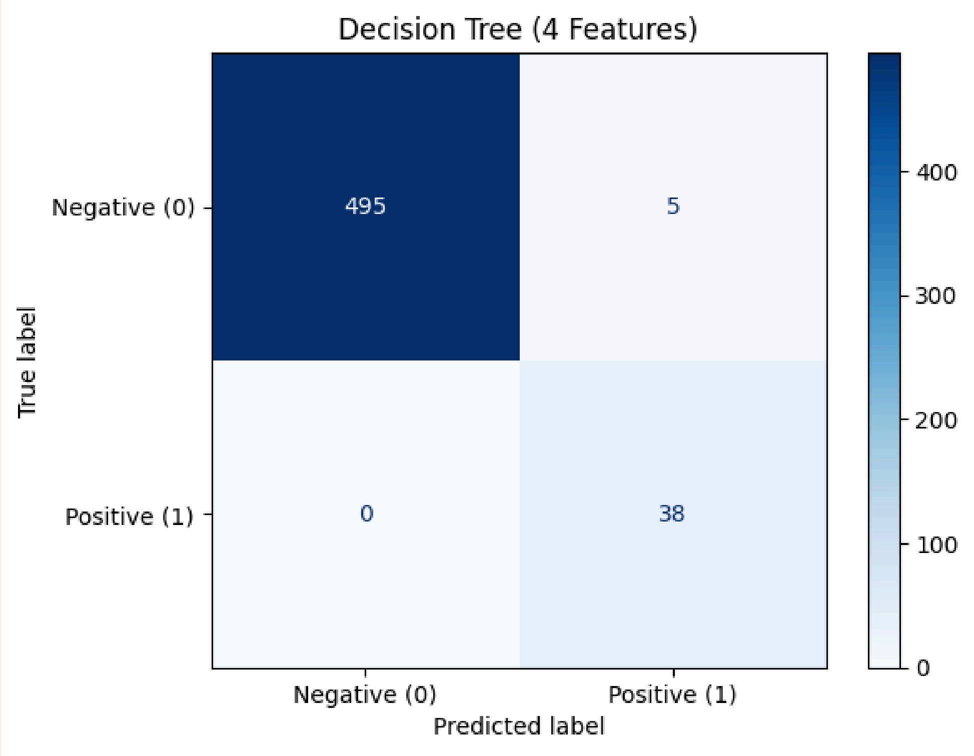


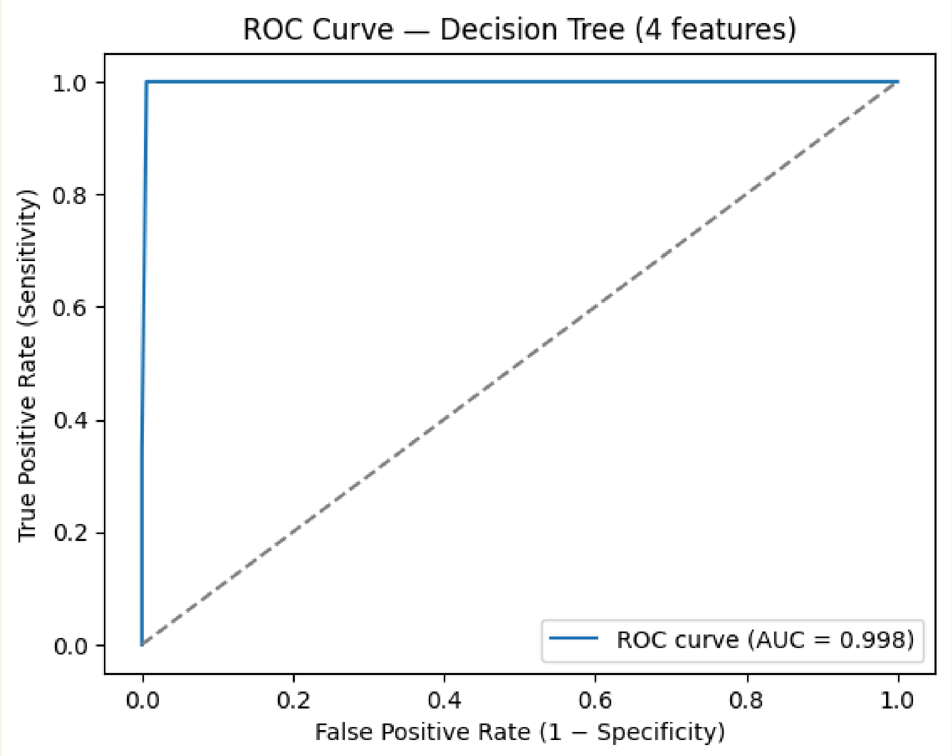


Model: Decision Trees









Trained on all features

Found top 4 features

Trained new model

Compare models

FEAUTURE SELECTION: DT

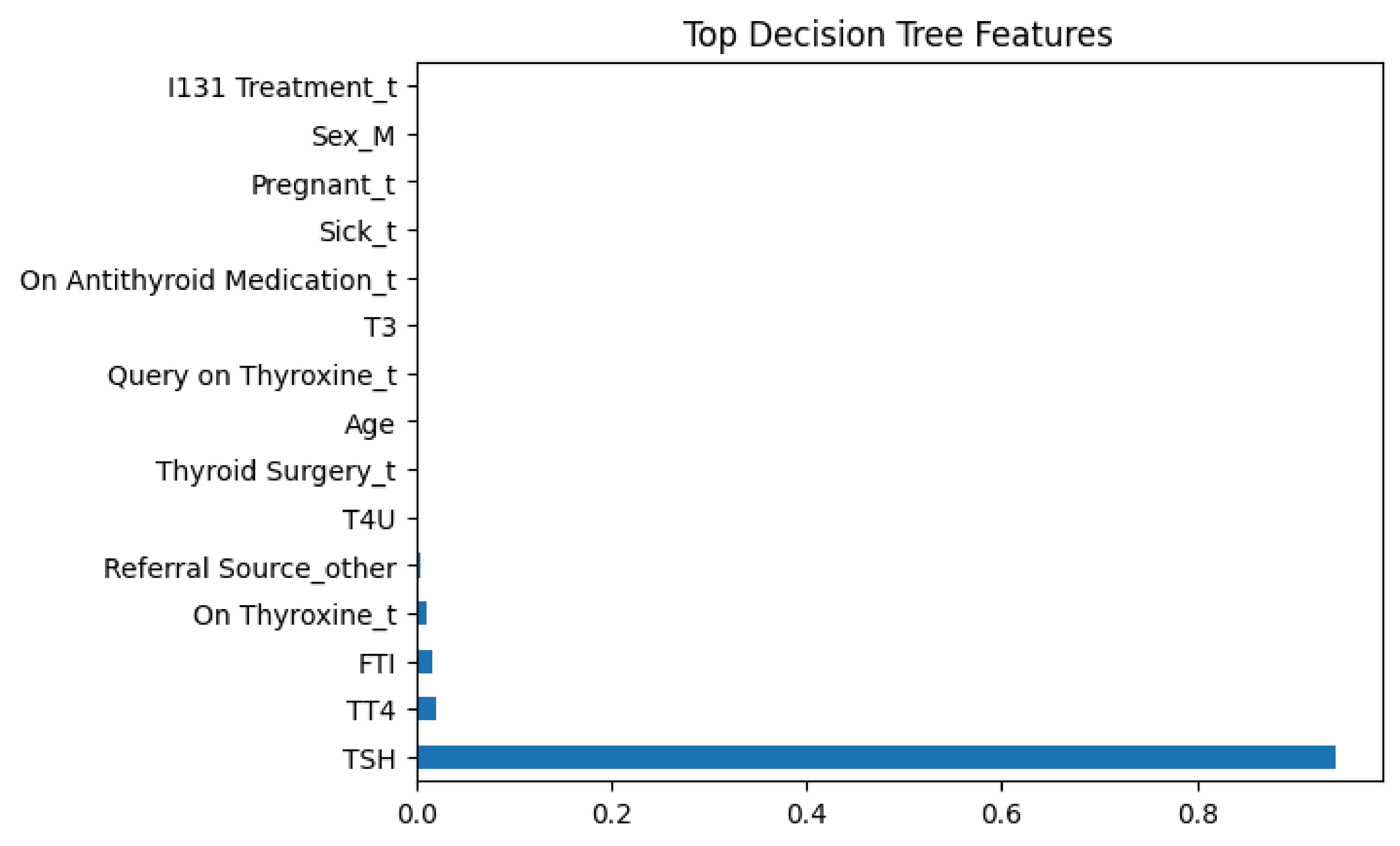
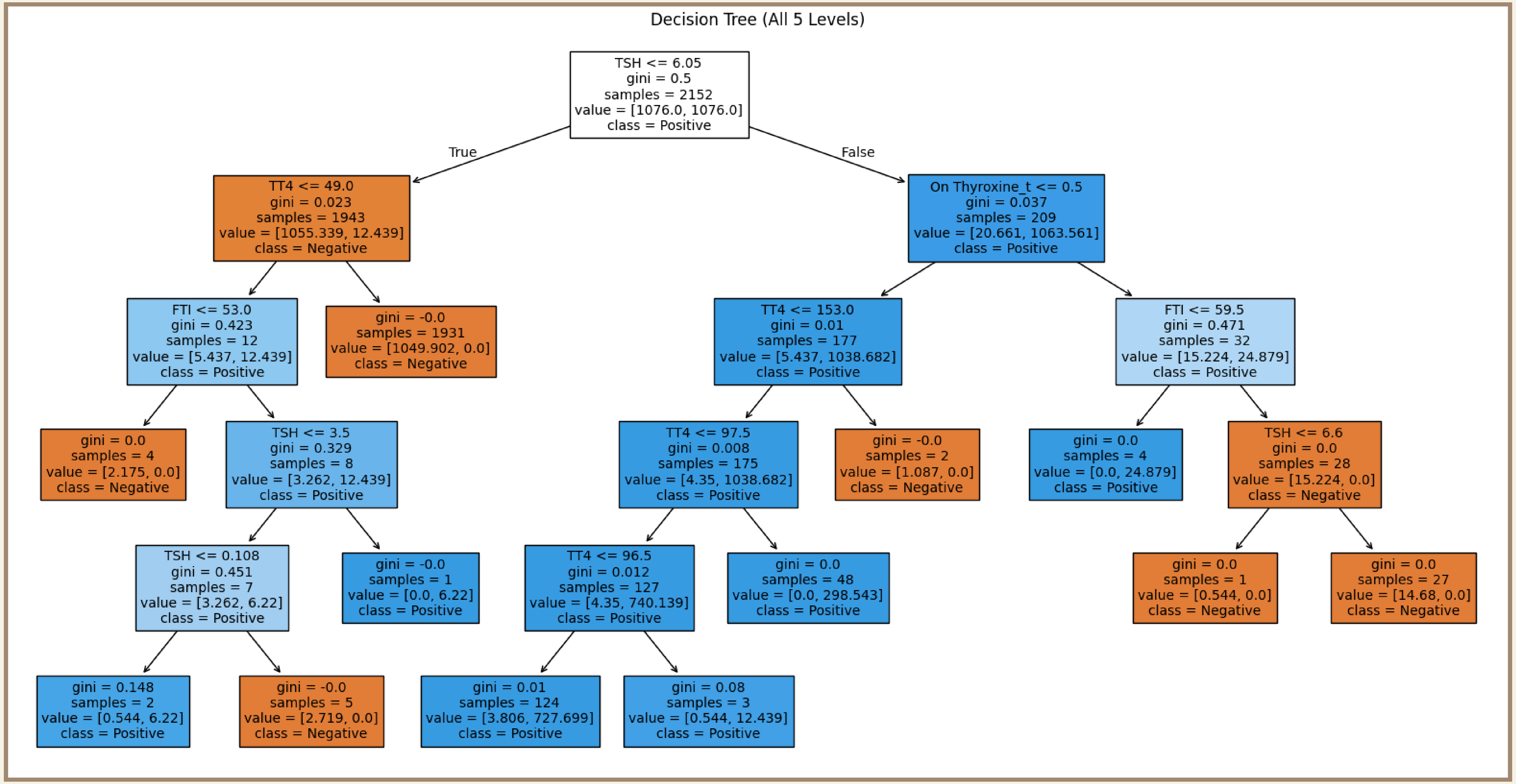


fig 1. feature importances from initial DT model

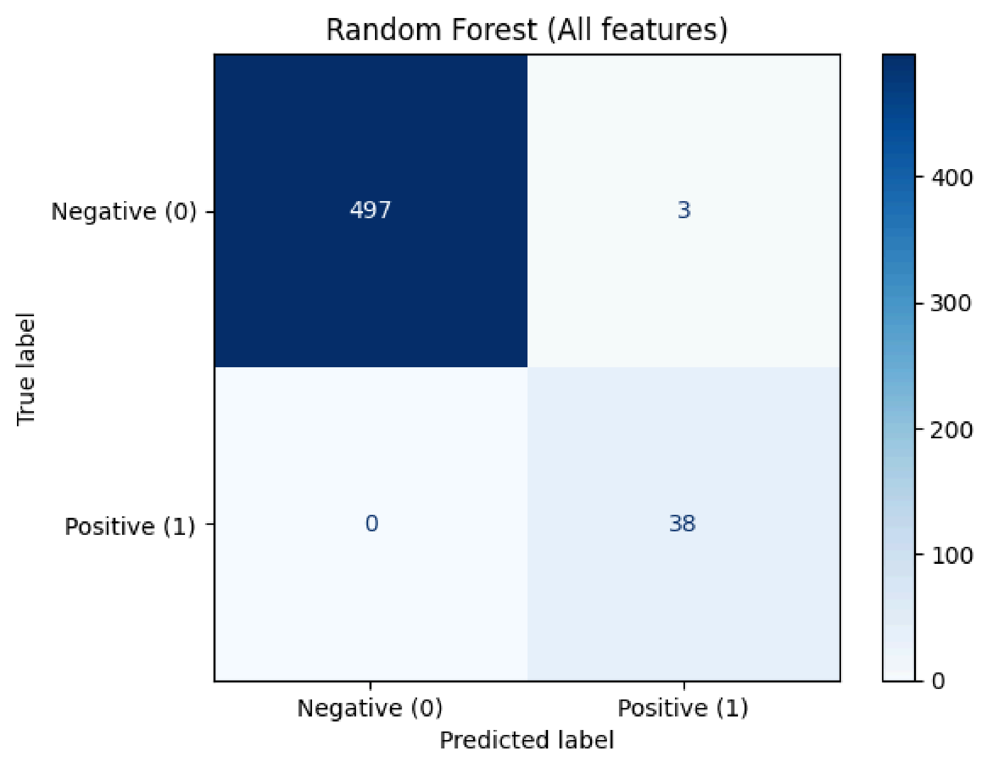
Top 4 Features:

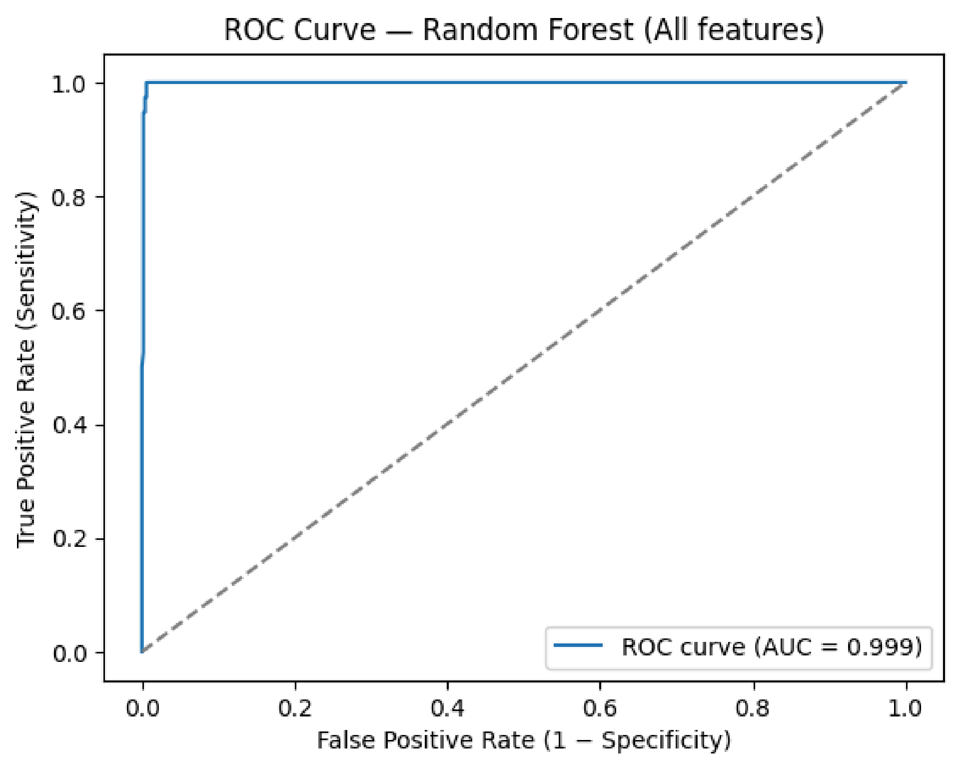
- TSH: Thyroid-Stimulating Hormone
- TT4: Total Thyroxine
- FTI: Free Thyroxine Index
- On Thyroxine_t

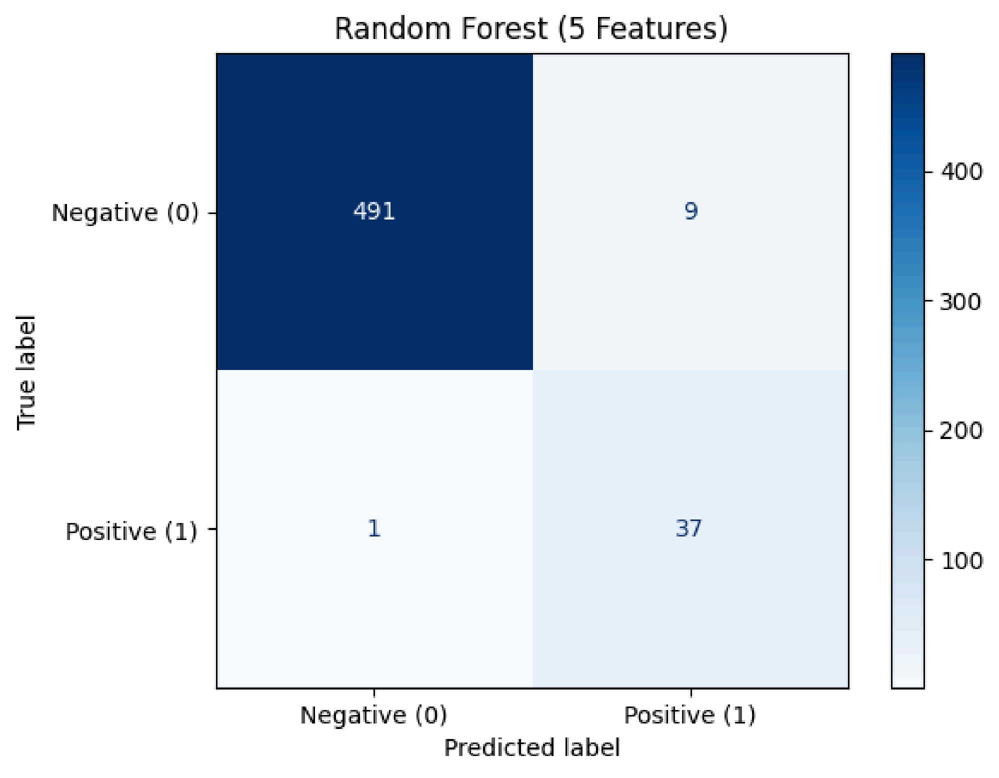
Model: Decision Tree

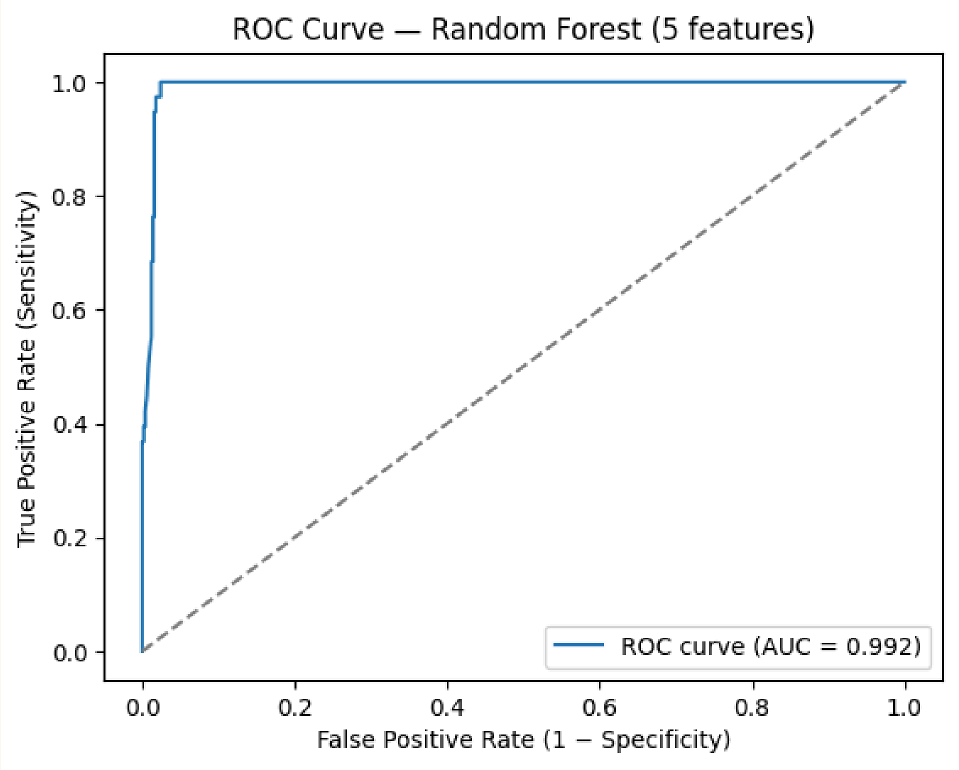


Model: Random Forest









Trained on all features

Found top 5 features

Trained new model

Compare models

FEATURE SELECTION: RF

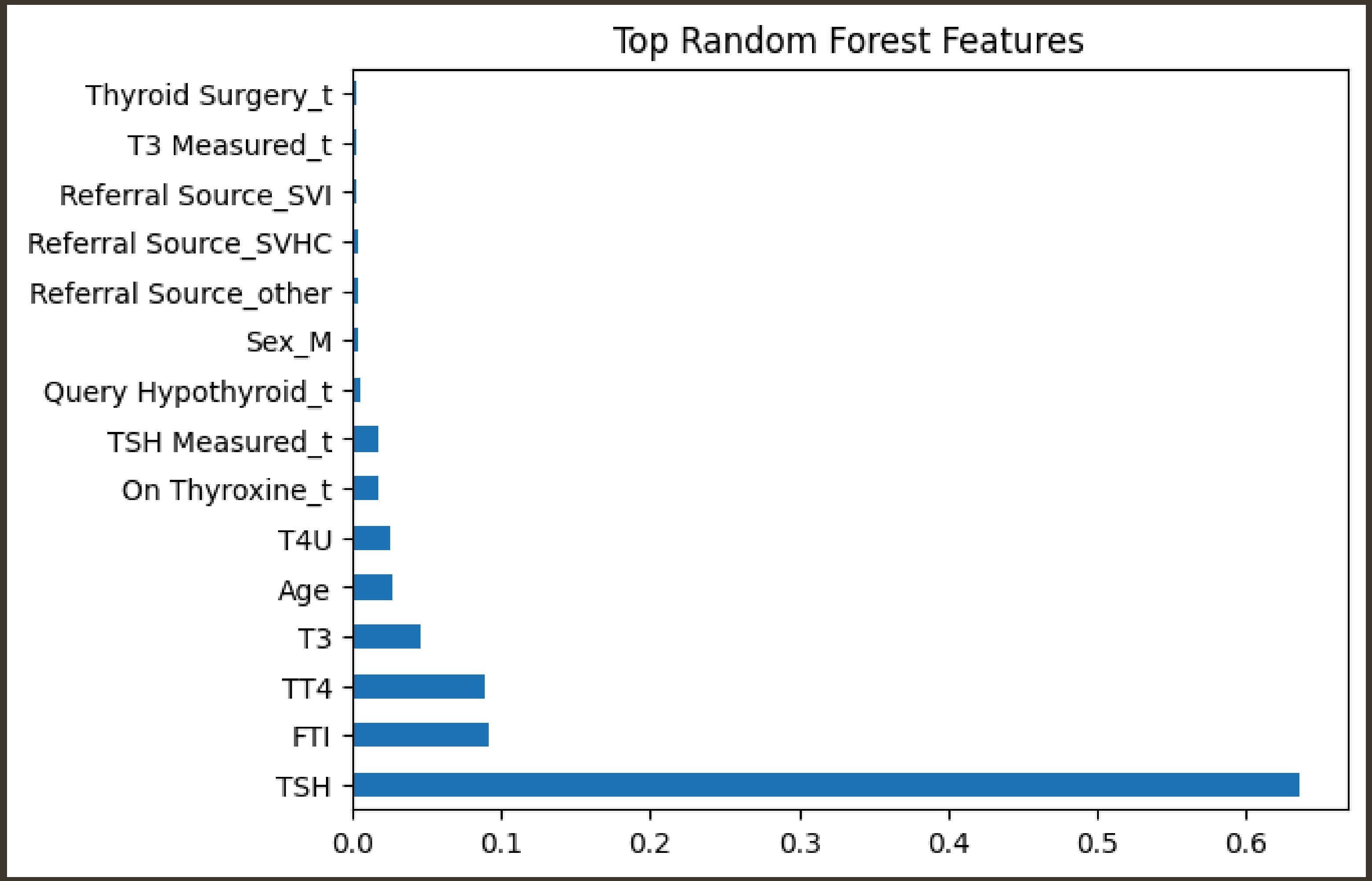


fig 1. feature importances from initial RF model

Top 5 Features:

- TSH: Thyroid-Stimulating Hormone
- FTI: Free Thyroxine Index
- TT4: Total Thyroxine
- T3: Triiodothyronine
- Age

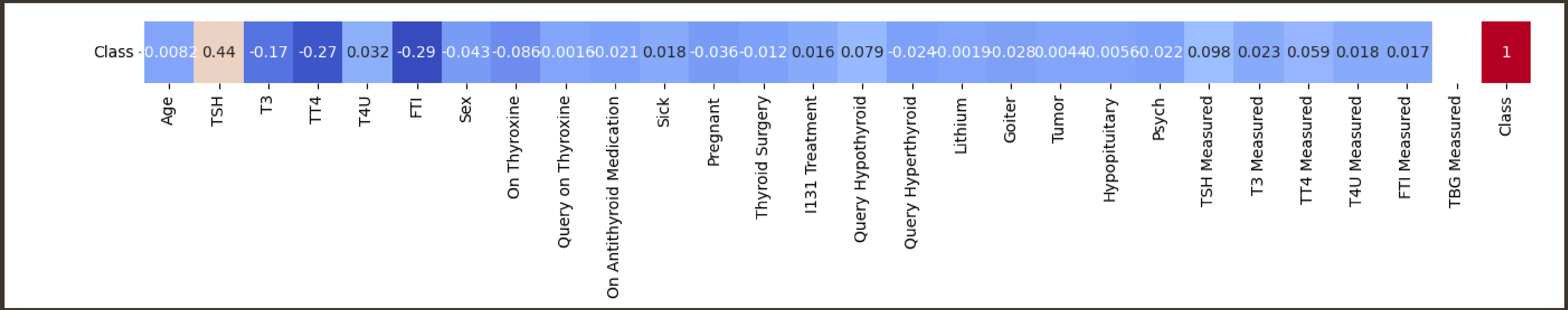
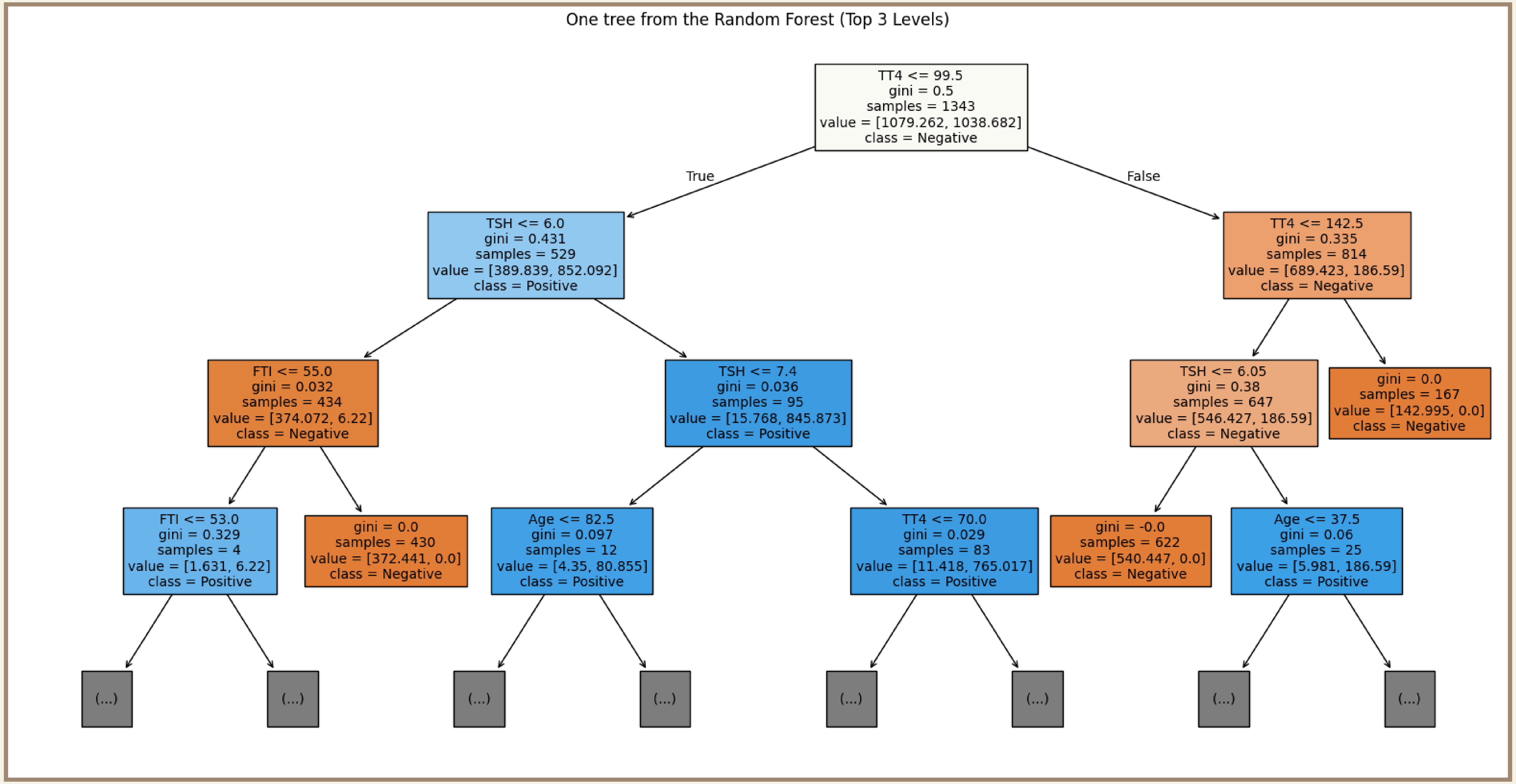


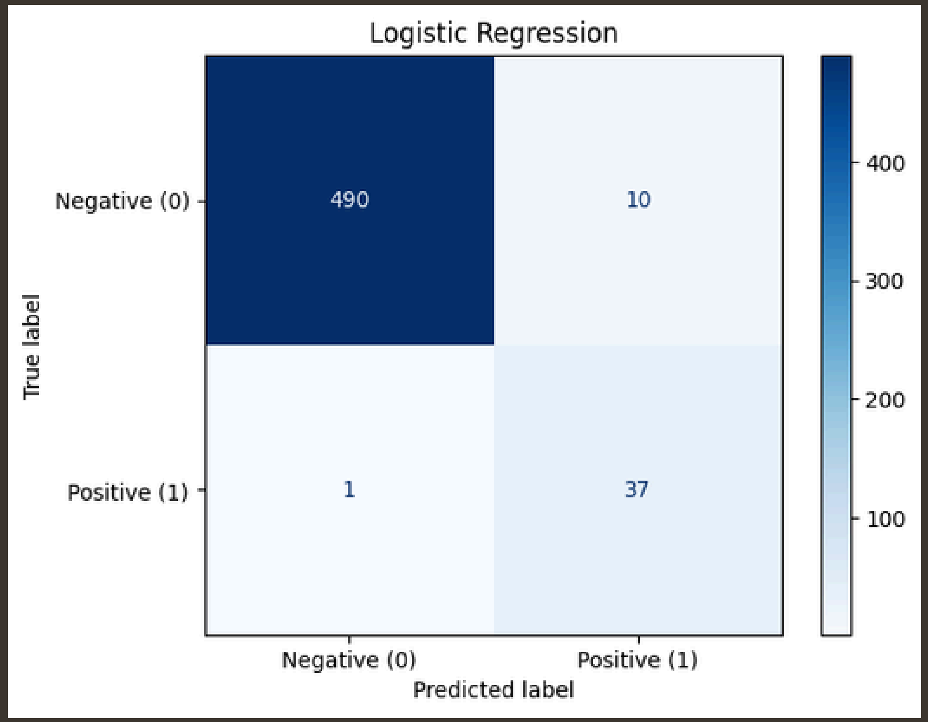
fig 2. snippet from heatmap of data

Model: Random Forest

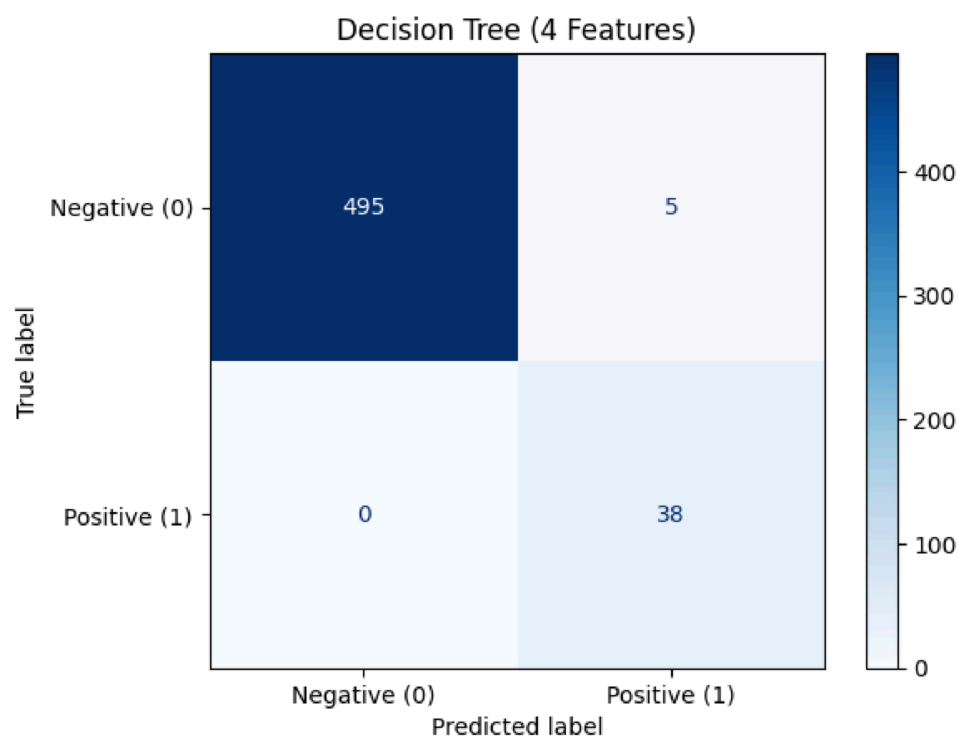


RESULTS

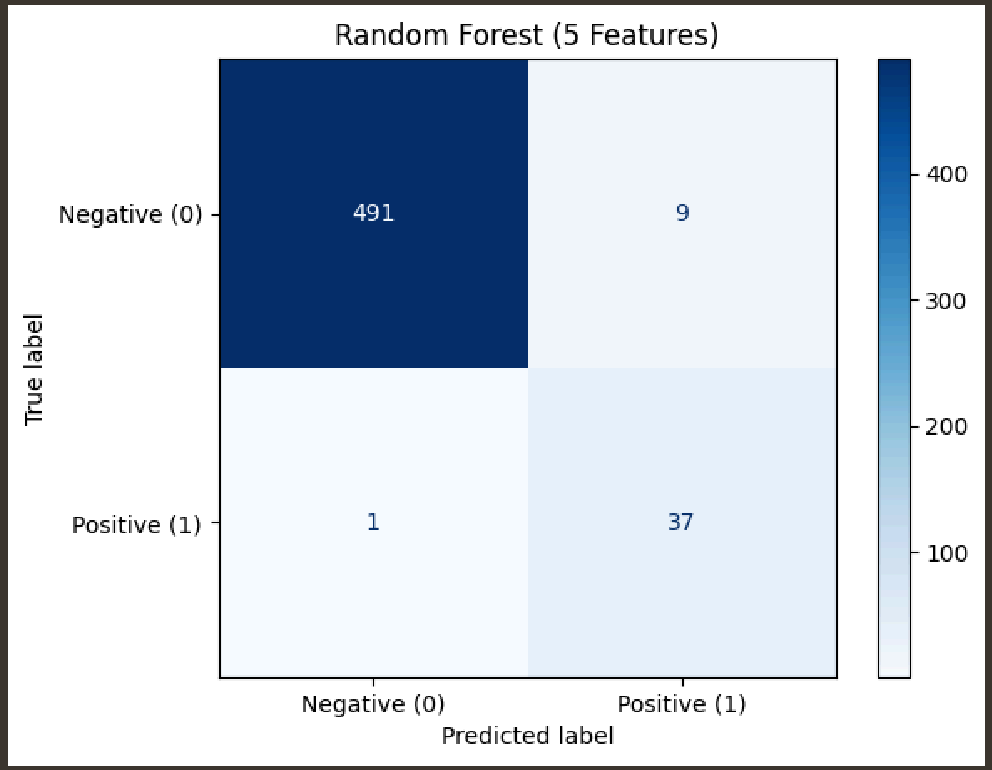
Zero Rule: accuracy = 92.9% (0% sensitivity)



Accuracy: 97.9%
Sensitivity: 97.4%
Specificity: 98%



Accuracy: 99.1%
Sensitivity: 100%
Specificity: 99%



Accuracy: 98.1%
Sensitivity: 97.4%
Specificity: 98.2%

Interpretation/discussion

- All models had very similar results
- Final Model: Random Forest
  - DT has a higher likelihood of having overfitted data
  - Doesn’t assume shape (linearity, etc.)
  - Thyroid behavior related to threshold (hormone expression > x); easily represented in RF
- Important features from RF and DT models: TSH, FTI and TT4
- Future data collection + diagnosis should focus on those features

LIMITATIONS

- **Imbalanced data at only 7% positive (38/538).**
- **Single source data.**
- **Simplified the hypothyroidism categorical class to binary.**

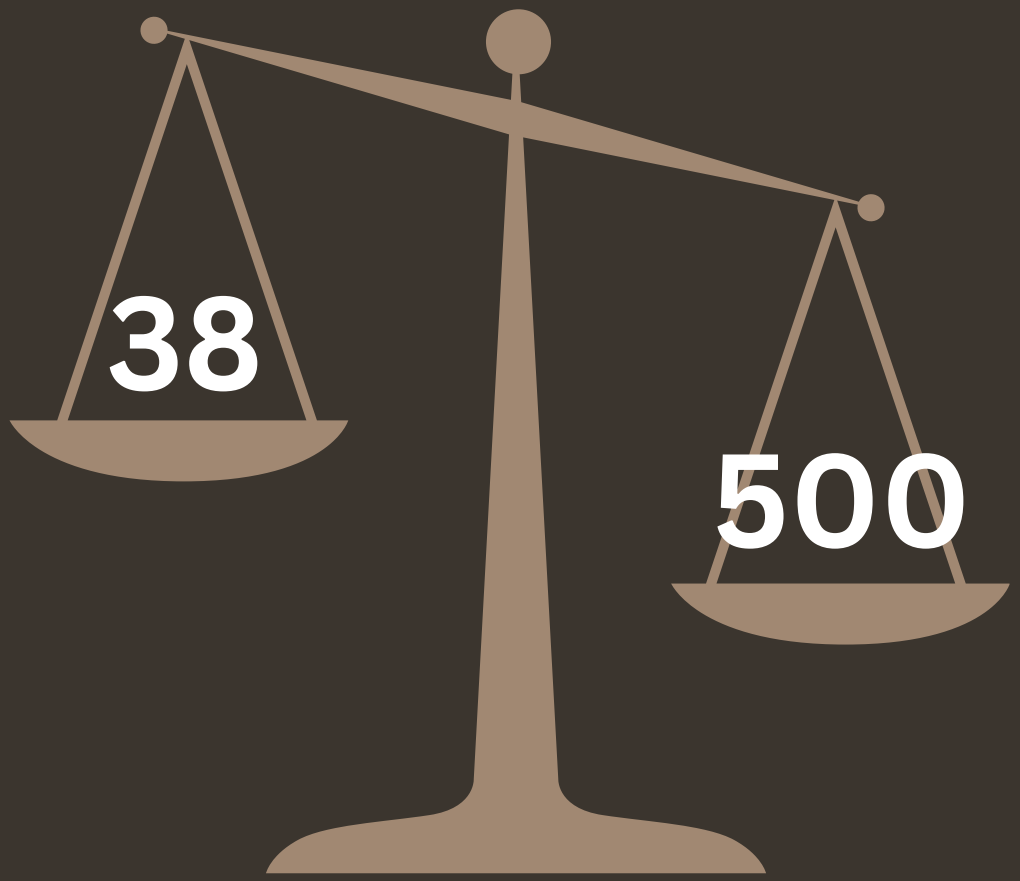
```
train_data["Class"] = train_data["Class"]. apply(lambda x: 0 if x == "negative." else 1 )
```

- **Used median scores in place of missing values.**

```
for col in numeric_cols:
  train_data[col] = train_data[col].fillna(train_data[col].median())
```

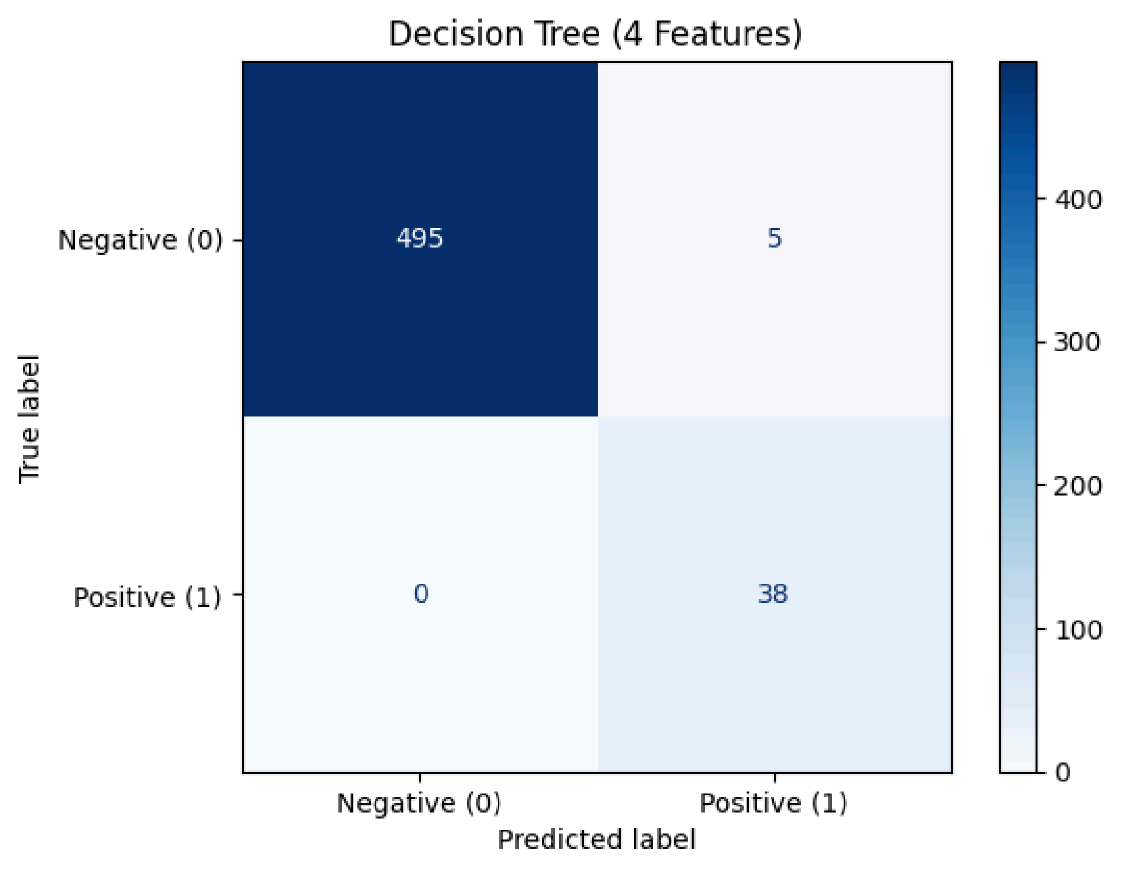
- **Removed records missing sex.**

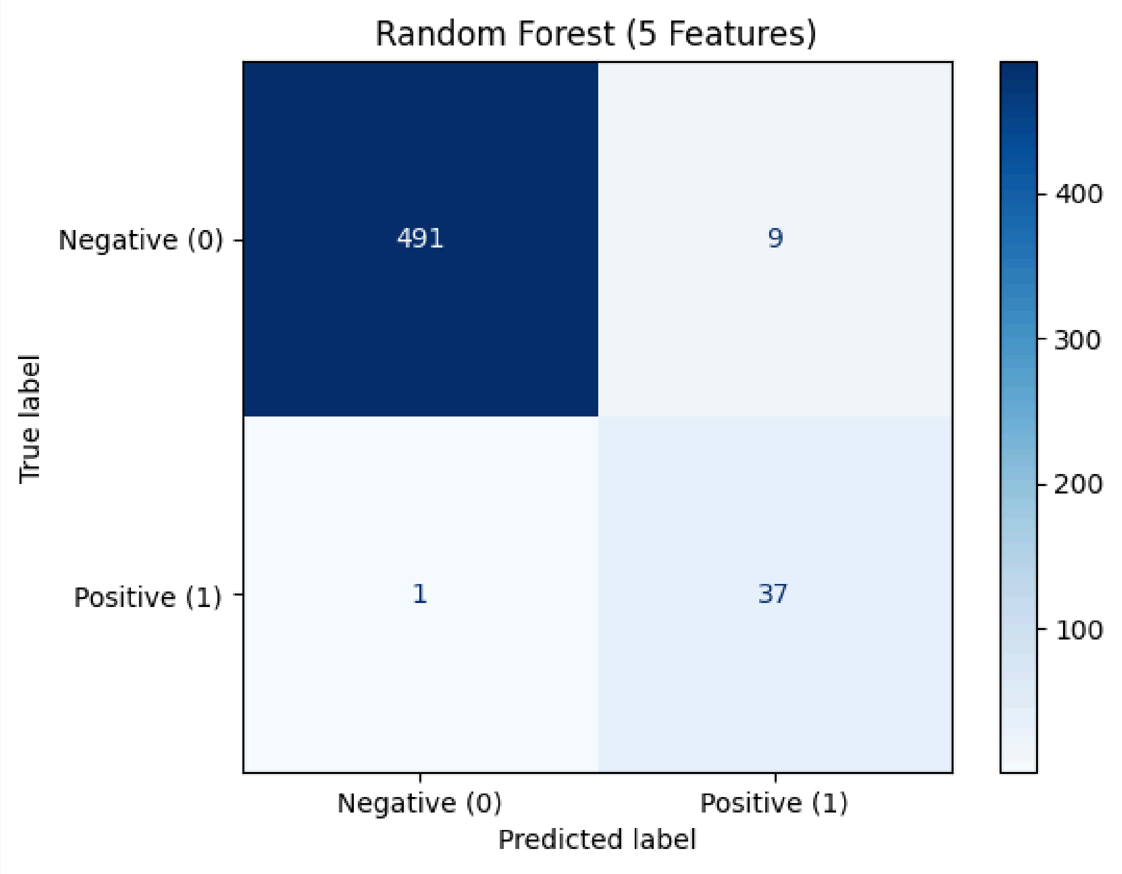
```
train_data = train_data.dropna(subset = ["Sex"])
```



Conclusion

Can we accurately predict whether a patient has hypothyroidism using a supervised model trained on a minimal set of features?





We can accurately predict whether a patient has hypothyroidism using a supervised model trained on a minimal set of features, as both the decision tree model and the random forest model were able to yield high levels of accuracy and sensitivity on minimal features, mainly focusing on TSH, FTI, and TT4 data.